

Reverse-Engineering Cupid’s Arrow:

The Neuroendocrinology of Falling in Love and the Feasibility
of a Pharmacobehavioural “Bonding Amplifier”

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Abstract

The love potion — Cupid’s arrow, the philtre of Tristan and Isolde — is among the oldest fantasies in human culture: a substance that compels another person to fall in love. We ask whether any real-world analogue could exist, and under what conditions. We proceed in two movements. First we ask the prior question the myth presupposes: *what is falling in love, chemically, and how is it produced?* Synthesising neuroimaging, endocrine and animal-model evidence, we describe romantic love as a goal-directed motivational state built from a dopaminergic “wanting” system (ventral tegmental area and caudate), an oxytocin/vasopressin plasticity system that consolidates a *partner-specific* preference, and a supporting endocrine cast (serotonin, cortisol, nerve growth factor, sex steroids), shaped by an evolutionary logic of biparental care and pair-bond commitment. A crucial principle emerges: the pair bond is a *conditioned* preference — neuromodulators supply reward and plasticity, but only repeated pairing with one individual supplies the “who.” We then test whether a periodically administered potion could amplify this process. We build a computational model coupling one-compartment pharmacokinetics, dopamine reward-prediction-error (temporal-difference) learning, oxytocin-gated plasticity, tolerance, extinction, mere-exposure and a baseline-compatibility term, and we sweep dosing frequency, pairing contingency, baseline compatibility, composition and dose timing. The model reproduces the literature’s constraints and yields a clear verdict. A single-dose, will-overriding “magic” potion is biologically incoherent: it has no way to specify a target, cannot invert aversion, and is defeated by pharmacokinetics. But a *combined oxytocinergic-dopaminergic amplifier, dosed a few times per week, correctly timed, and paired with genuine positive interaction with one person*, raises the modelled probability of pair-bond formation from near zero (placebo) to ~ 0.73 over eight weeks — provided three conditions hold: accurate aiming (contingent pairing), a non-negative pre-existing compatibility, and managed dosing to avoid tolerance and over-exposure. The modern Cupid’s arrow is therefore not a compulsion but an *amplifier of a spark that must already exist*; fittingly, like the myth, the arrow still has to be aimed. We close with a concrete proposed formulation, the necessary conditions, and the consent-centred ethics without which such a tool would be indefensible.

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1 Introduction

Across cultures and millennia, humans have imagined that love could be *administered*. Cupid looses an arrow and Apollo burns for Daphne; Isolde and Tristan drink the philtre and are bound past reason; Shakespeare’s Oberon squeezes a flower’s juice on sleeping eyes. The fantasy is remarkably specific: a discrete intervention, delivered to a target, that produces involuntary, durable, person-directed love. This paper takes that fantasy seriously enough to ask whether science permits any version of it, and answers in a deliberately constructive spirit: not “is the fairy tale literally true?” (it is not) but “what is the strongest real protocol the biology of love would actually support, and what conditions must it satisfy?”

The question is timely rather than whimsical. The last three decades have produced a detailed, replicated neurobiology of romantic attachment; intranasal oxytocin has been trialled as a “relationship enhancer”; MDMA-assisted couples therapy has reached proof-of-concept; and a serious bioethics of “love drugs” has emerged [22, 23, 68]. The ingredients of a real intervention exist in the literature. What has not been done is to assemble them into an explicit feasibility argument and ask, quantitatively, how far they can go.

We organise the paper around a logical ordering that the myth itself implies. A potion can only “cause love” if we first know what love *is* and how it is normally produced. We therefore ask the constitutive question before the interventional one:

Part I. What is falling in love, chemically? (Sections 2–2.5)

Part II. Why does it happen — the evolutionary logic? (Section 3)

Part III. How is it triggered and amplified — the “recipe”? (Section 4)

Part IV. Can a potion amplify the recipe — a computational model. (Section 5–6)

Part V. The verdict, a concrete formulation, and the ethics. (Sections 7–8)

Contribution and stance. Our central thesis is that the literal love potion is impossible for principled biological reasons, but that a *periodically administered bonding amplifier* is feasible in principle and would substantially raise the odds of love forming — under strict conditions that the same biology dictates. We support this with (i) a synthesis of the verified primary literature and (ii) an explicit, reproducible computational model that turns the qualitative argument into dose–response, contingency and compatibility curves. Throughout we flag contested findings honestly: the intranasal-oxytocin literature has a documented replication problem [43, 55, 64], several endocrine results rest on small samples, and “human pheromones” are, on present evidence, not established [69]. We are careful to distinguish what is robust from what is suggestive, because a feasibility claim is only as honest as its weakest link.

2 Part I — What falling in love is, chemically

2.1 Three systems: lust, attraction, attachment

The most useful organising framework partitions mating-related emotion into three dissociable but interacting neuroendocrine systems [26, 27]. *Lust* (the sex drive) is driven primarily by gonadal steroids — testosterone and estrogens — and is relatively undirected. *Attraction* (romantic love proper) is a focused, high-energy, intrusive state associated with elevated central dopamine and, hypothetically, noradrenaline, and possibly lowered serotonin; it directs courtship energy toward one preferred individual. *Attachment* (the calm security of an established pair bond) is associated with the neuropeptides oxytocin and vasopressin. The three systems can operate semi-independently — one can feel attachment without lust, or lust without attraction — which is why “love” is not a single molecule but a layered process. This dissociation is the first warning to any would-be potion-maker: there is no master switch.

2.2 The dopaminergic “attraction” system: wanting a specific person

Functional neuroimaging consistently shows that viewing a beloved’s face, relative to a familiar non-loved face, activates dopamine-rich reward and motivation regions. Early studies localised activation to the caudate nucleus, putamen, insula and anterior cingulate, with deactivation of amygdala and social-judgement cortex [7, 8]; the defining study added the right ventral tegmental area (VTA) and medial caudate, and — crucially — showed that caudate activity scaled with scores on the Passionate Love Scale and VTA activity with rated facial attractiveness [5]. The authors concluded that early romantic love is best understood not as an emotion but as a *goal-directed motivation system* — “a neural mechanism for mate choice” [28]. Remarkably, the same VTA/striatal engagement persists in people still intensely in love after some 21 years of marriage [1], so the reward signature is not merely a transient of infatuation.

The mechanistic interpretation matters for our question. In the incentive-salience framework, mesolimbic dopamine encodes “wanting” (the motivational pull toward a stimulus), which is dissociable from “liking” (its hedonic impact) [10]. Romantic love is, in this reading, intense *wanting* of a particular person; and because dopaminergic wanting can be sensitised and exhibits tolerance and withdrawal, love legitimately shares machinery with addiction [14, 29]. This is the kernel of optimism for a potion — the attraction system is a reward-learning system, and reward learning can be pharmacologically influenced — but also the kernel of its central difficulty, addressed in Section 4: dopamine encodes *how much* you want, not *whom*.

2.3 Oxytocin and vasopressin: the plasticity that fixes the target

If dopamine supplies the motivational charge, the neuropeptides supply the *specificity* — they help bind that charge to one individual. The decisive evidence comes from microtine voles. In the monogamous prairie vole, central oxytocin induces partner preference in females even without mating, and an oxytocin antagonist blocks mating-induced preference; vasopressin plays the homologous role in males [39]. Partner preference can be conferred by over-expressing the vasopressin V1a receptor in the ventral forebrain [56], and transferring a single V1a-receptor gene into the *promiscuous* meadow vole is sufficient to induce prairie-vole-like bonding [44]. The lesson, formalised by Young and Wang [70], is that it is the *distribution of receptors*, not the peptide itself, that gates the capacity to bond, and that a bond forms when oxytocin/vasopressin (carrying the partner’s social-recognition cues) and mesolimbic dopamine (carrying reward) are *co-activated* in the nucleus accumbens during cohabitation. Bonding requires *simultaneous* engagement of oxytocin and D2-type dopamine receptors in the accumbens: blocking either one abolishes the preference induced by an agonist of the other [47]. Dopamine receptor subtypes play opposing roles — accumbens D2 activation facilitates bond *formation* while D1 activation opposes it, and a post-bonding switch toward D1 signalling underlies mate-guarding [2, 33].

In humans the evidence is real but far weaker, and must be reported with its caveats. A single intranasal dose of oxytocin increased trust in an economic game [42]; in couples, oxytocin improved conflict communication and lowered cortisol [18]; and, most relevant here, oxytocin selectively increased nucleus-accumbens/VTA reward responses to a *partner’s* face but not to other familiar women [60], a plausible human echo of the vole reward-conditioning mechanism. However, the broader intranasal-oxytocin literature is statistically underpowered and replicates poorly; the trust effect in particular has not held up [55, 64]. We return in Section 4.2 to the still sharper *pharmacological* problem — that very little intranasally administered oxytocin reaches the brain [43].

2.4 The supporting endocrine cast

Beyond the two lead systems, falling in love leaves a measurable, if smaller and more variable, endocrine signature. Early-stage lovers show platelet serotonin-transporter densities reduced by ~40%, statistically indistinguishable from untreated obsessive-compulsive patients, a correlate of the intrusive, preoccupied quality of new love [51]. New lovers show altered hormone profiles — raised cortisol and a blunting of the normal sex difference in testosterone (it falls in men and rises in women) — that *normalise* when re-tested 12–24 months later [50]. Plasma nerve growth factor is elevated in proportion to passion intensity and likewise normalises over the same window [24]. The cortisol story is genuinely unsettled: acute reflection on a partner *raises* salivary cortisol [49], yet integrated daily cortisol output is *lower* in new lovers [66] — a contradiction best read as love being an acute arousal superimposed on a generally calmer baseline. Finally, the best-powered evidence in this cluster, a meta-analysis of 66 studies, confirms that pair-bonded men and fathers carry lower testosterone, consistent with a hormonal shift away from mating effort toward bonding/parenting [34]. Two integrative points stand out for the potion question: this endocrine signature is *transient* (it indexes the passionate phase, not the durable bond [11]) and it is *downstream* — a consequence of being in love, not an obvious lever for causing it.

2.5 Pheromones and body odour: a contested, minor adjunct

Folk theories of attraction lean heavily on smell, and the commercial market sells “pheromone” products outright; intellectual honesty requires separating the small kernel of real science from this. The careful verdict is sceptical. Humans possess a vestigial vomeronasal organ, and the genes essential to vomeronasal pheromone transduction — *TRPC2* and the *V1R* receptor family — are pseudogenised in humans and other catarrhine primates [45, 72]. The four

steroids marketed as “human pheromones” (androstadienone, estratetraenol, androstenone, androstenol) were never identified by the rigorous bioassay-led method used in other species, and the behavioural claims fail well-powered replication: exposure to androstadienone or estratetraenol does not reliably alter attractiveness ratings or related judgements [35, 69]. What *is* real is narrower: neonates respond to secretions of maternal areolar glands — the best current candidate for a genuine human chemosignal [19] — and body-odour preferences linked to MHC/HLA genotype may bias attraction, although even this is population-dependent and contested [16, 17, 65]. For our purposes the implication is precise: olfactory cues can at most nudge *baseline compatibility* (the κ parameter of Section 5) through familiarity and comfort; they are not a plausible active principle for a love potion, and any formulation that claims otherwise is selling folklore.

3 Part II — Why it happens: the evolutionary logic

Romantic love looks engineered because it was — by natural selection, for a function. Human infants are extravagantly altricial and costly, requiring years of provisioning and protection; this favoured biparental care and, with it, a mechanism to keep two adults invested in one another long enough to raise a child [38, 63]. On the leading synthesis, romantic love is precisely that mechanism: a *commitment device* that focuses mating effort on a single partner and motivates the maintenance of the bond against the pull of alternatives [30]. The adult pair bond co-opts the older infant-caregiver attachment system — adult romantic love was reconceived as an attachment process built on the same biosocial machinery [13, 36] — which is why romantic and maternal love share reward/peptide circuitry in the brain [8].

Two further observations matter for feasibility. First, mate choice is not random: it is shaped by cues of compatibility and quality — assortative preferences, facial and symmetry cues, and (contested) odour/MHC signals [46, 65] — which means a real target carries a *prior valence* that an intervention must work with or against. Second, romantic love is a near-human-universal: ethnographic survey finds evidence of it in roughly 89% of sampled cultures [40], consistent with an evolved adaptation rather than a cultural invention. The evolutionary reading delivers a constraint we will encode directly into the model: a system built to bond an individual to a *suitable* partner is not built to be hijacked toward an arbitrary one. Love evolved to be *selective*; a potion that ignores selection is fighting the function of the very system it hopes to recruit.

4 Part III — How love is triggered and amplified: the recipe

We can now state, as an explicit “recipe,” the conditions under which the systems of Part I actually produce a pair bond. This synthesis is the bridge to the potion question: a potion can only help by supplying or strengthening one of these ingredients.

The four necessary ingredients.

1. **Reward / “wanting”** — a dopaminergic signal that tags encounters with the person as rewarding and motivating [5, 10].
2. **A plasticity gate** — oxytocin/vasopressin co-activation that consolidates the encounter into a durable, *partner-specific* preference [47, 70].
3. **Contingent pairing (the “aim”)** — repeated temporal/attentional binding of that reward to *one* individual, so that a non-specific internal state becomes attached to a specific external cue. This is associative learning, formally a conditioned partner preference [2, 70].

4. **Non-negative baseline compatibility** — a target whose intrinsic valence is not strongly aversive, since the systems above amplify and consolidate an existing signal rather than manufacture one from nothing [30].

4.1 Behavioural triggers: how humans actually supply the aim

Social psychology has, in effect, catalogued the delivery system for ingredient 3 — procedures that reliably bind arousal, reward and familiarity to a particular person. *Misattribution of arousal* turns generic physiological activation into attraction when an attractive target is the salient explanation: men aroused by a fear-inducing bridge were markedly more drawn to a woman they met there [21], an effect grounded in two-factor theory [59] and shown to be real but moderate ($d \approx 0.27$) and strongest when the arousal source is ambiguous [31]. *Mere exposure* and *propinquity* convert repetition into liking [25, 54, 71], although the effect is curvilinear — over-exposure breeds boredom [12], a fact we encode as an over-dosing penalty in the model. *Escalating reciprocal self-disclosure* — the “36 questions” — generates closeness in 45 minutes regardless of prior matching [3]; *interpersonal synchrony* delivers affiliation to the specific person one moves in time with [37, 52, 67]; *shared novel and arousing activities* channel excitation into an existing relationship [4]; and similarity and reciprocated liking add further pull [6, 15]. The common thread is contingency: each method temporally and attentionally binds an internal state to one person. *This is the aim of the arrow*, and — importantly — it is achievable without any drug at all.

4.2 Pharmacological levers — and their hard limits

What can chemistry add? Drugs can plausibly amplify ingredients 1 and 2 (reward and the plasticity gate) and lower the fear/defensiveness that opposes them. MDMA raises oxytocin and produces prosocial, empathogenic openness — with the size of the prosocial effect tracking oxytocin more closely than blood MDMA [20] — and MDMA-assisted couples therapy has reached proof-of-concept (uncontrolled, $n = 6$ couples) as a drug that lowers defensiveness while the relational work supplies specificity [41, 53]. Intranasal oxytocin shifts real couple behaviour [18] and partner-specific reward [60].

But the limits are severe and define the whole feasibility question:

- **Oxytocin barely reaches the brain.** Its plasma half-life is only $\sim 3\text{--}5$ minutes, it crosses the blood–brain barrier poorly, and on pooled human data at most $\sim 0.005\%$ of an intranasal dose reaches cerebrospinal fluid; central levels, when they rise at all, lag plasma by ~ 75 minutes and are uncorrelated with it [43, 62]. Whether direct nose-to-brain transport rescues the route is genuinely unresolved [58].
- **Dopamine cannot be “given.”** Dopamine does not cross the blood–brain barrier; only precursors (L-DOPA) or releasers (amphetamine) act centrally, and they act *non-specifically* across the whole brain [32].
- **A blunt dopamine flood can *prevent* bonding.** In prairie voles, amphetamine — maximal, non-specific accumbens dopamine — *impairs* pair-bond formation rather than causing it, because it hijacks the very reward signal the bond needs but decouples it from the partner [48]. More reward chemistry is not more love.
- **Tolerance.** Repeated dopaminergic stimulation desensitises receptors on a minutes-to-hours timescale [9, 14], capping how hard the reward lever can be pushed.
- **Dose is non-monotonic.** For intranasal oxytocin, lower doses (~ 8 IU) can exceed higher ones (~ 24 IU), and the central effect follows an inverted-U [57, 58].

The synthesis is unambiguous and, for the romantic, oddly hopeful in its shape. Pharmacology can supply and amplify reward and plasticity; it cannot supply specificity. *Specificity must come from conditioning* — from pairing the amplified state with one person over time [70]. A potion is therefore conceivable only as an *amplifier of a paired, repeated, mutually positive courtship*, never as a stand-alone compulsion. The rest of the paper quantifies exactly how much such an amplifier could do.

5 Part IV — A computational feasibility model of the arrow

5.1 Rationale

The verbal argument above makes qualitative predictions: that a combined amplifier should beat either ingredient alone, that periodic dosing should help but with diminishing and then negative returns, that contingency should be necessary, that compatibility should gate the effect, and that timing should matter. We formalise these in a single model so the predictions can be made quantitative and their interactions examined. The model deliberately encodes the constraints of Section 4.2; it is calibrated to the *order of magnitude* of the cited literature rather than fitted to human bonding data (no such dataset exists), and is therefore a *qualitative feasibility and sensitivity tool*, not a clinical predictor. We state this limitation plainly and return to it in Section 8.

5.2 Model definition

We model the growth of a target-specific bond strength $B_n \in [0, 1]$ over a sequence of n *encounters* with one person, some of which are augmented by a dose of the potion and paired with positive interaction. The learning rule is temporal-difference reward learning [61]: the person is the conditioned stimulus, the current bond strength is the expected value, and each encounter yields a reward whose prediction error updates the bond. The oxytocinergic component gates the learning rate (plasticity); the dopaminergic component scales the reward magnitude; specificity enters as a contingency factor c .

At encounter n the central drug effects are

$$O_n = g_{OT}(d_{OT}) \phi_{OT}, \quad D_n = \frac{d_{DA}/(\frac{1}{2} + d_{DA})}{1 + T_n} \phi_{DA}, \quad (1)$$

where $g_{OT}(\cdot)$ is an inverted-U (log-Gaussian) dose-response peaking at a low-moderate dose [57, 58], d_{OT}, d_{DA} are the oxytocin and dopaminergic doses, $\phi_{OT}, \phi_{DA} \in [0, 1]$ are dose-encounter timing overlaps (the central oxytocin peak lags the dose by ~ 75 min, so the encounter must be scheduled to coincide; 62), and T_n is accumulated dopaminergic tolerance [9]. The encounter reward is

$$R_n = \kappa + \beta(1 - e^{-n/n_0}) - \rho(f - 3)_+ + g_D D_n + \varepsilon_n - \underbrace{\eta D_n (1 - O_n c)}_{\text{ungated-flood penalty}}, \quad (2)$$

where κ is the target’s baseline compatibility (intrinsic valence), the second term is saturating mere-exposure familiarity [12, 71], the third is an over-exposure/boredom penalty for dosing frequency f above three per week [12], ε_n is Gaussian noise, and the final term encodes the finding that non-specific dopamine that is *not* gated by oxytocin and aimed by contingency can disrupt rather than build a bond [47, 48]. The oxytocin-gated, contingency-scaled learning rate and the TD update are

$$\alpha_n = \alpha_0 (1 + g_O O_n) c, \quad B_n = \left[B_{n-1} + \alpha_n (R_n - B_{n-1}) \right] e^{-\lambda \Delta t}, \quad (3)$$

Table 1: Principal model parameters, with the literature motivating each. Values are order-of-magnitude calibrations, not fitted constants. “Confidence” refers to how well the underlying quantity is established in humans.

Symbol	Meaning / literature anchor	Confidence
α_0	Baseline plasticity (learning rate)	assumption
g_O	Oxytocin gain on plasticity; OT gates accumbens consolidation [47, 70]	qualitative
g_D	Dopaminergic gain on reward magnitude [5, 10]	qualitative
$g_{OT}(\cdot)$	Inverted-U dose–response, peak at low dose [57]	high (shape)
ϕ_{OT}	Dose–encounter timing overlap; central OT peak ~ 75 min [62]	high
η	Ungated-flood penalty; amphetamine impairs bonding [48]	qualitative
β, n_0	Mere-exposure familiarity, saturating [12, 71]	moderate
ρ	Over-exposure/boredom penalty [12]	moderate
ξ, τ_T	Dopaminergic tolerance build/recovery [9]	moderate
λ	Extinction of bond without reinforcement	assumption
κ	Baseline compatibility (intrinsic target valence)	swept
c	Contingency / precision of pairing (the “aim”)	swept
θ	“In love” threshold on B	convention

with extinction λ acting over the inter-encounter interval Δt (set by the dosing frequency), and clipping to $[0, 1]$. Tolerance accrues with dopaminergic dosing and recovers between encounters, $T_n = (T_{n-1} + \xi D_n) e^{-\Delta t / \tau_T}$. A courtship “succeeds” if B exceeds a threshold $\theta = 0.6$ at the eight-week horizon; we report P_{bond} , the Monte-Carlo probability of success over 800 stochastic runs. Full equations and parameter values are in the accompanying code (`arrow_model.py`); key parameters are listed in Table 1.

6 Results

6.1 Periodic dosing works — but more is not better

Figure 1 shows P_{bond} as a function of how many augmented, paired encounters occur per week. The combined amplifier rises steeply from essentially zero at one encounter per week to $P_{\text{bond}} = 0.73$ at three per week — the cadence in the canonical “love potion” intuition — peaks at $P_{\text{bond}} = 0.81$ at five per week, and then *falls* to 0.62 at seven per week as dopaminergic tolerance and over-exposure boredom overtake the benefit of repetition. Periodic administration is thus genuinely effective, and there is an optimum: the model predicts an inverted-U in dosing frequency, the quantitative expression of “familiarity breeds contempt” meeting “tolerance.”

6.2 Neither ingredient alone suffices: the gated combination does

Figure 2a compares compositions at the headline protocol (three well-aimed encounters per week, contingency $c = 0.8$). Placebo, oxytocin-only and dopamine-only each fail to cross the threshold ($P_{\text{bond}} = 0.00$), even though they do raise mean bond strength partway (final $B = 0.24, 0.32, 0.41$ respectively). Only the *combined* amplifier crosses ($P_{\text{bond}} = 0.73$, final $B = 0.63$). This is the model’s expression of the obligatory oxytocin–dopamine co-activation that the vole accumbens requires [47, 70]: the dopaminergic component supplies reward magnitude, the oxytocinergic component supplies the plasticity to consolidate it, and falling in love is a threshold phenomenon that only their conjunction reliably reaches.

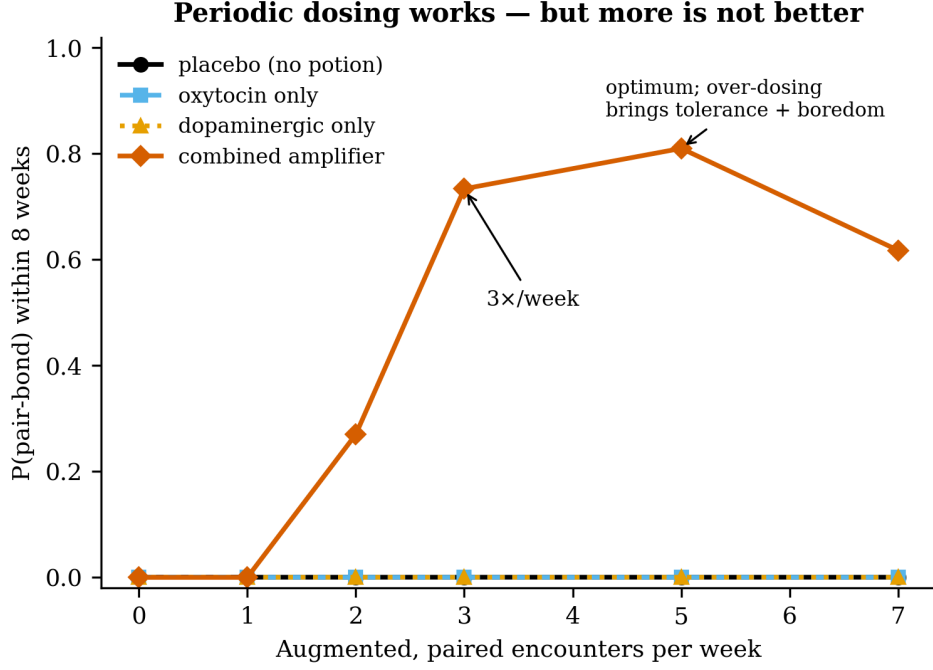


Figure 1: Probability of pair-bond formation within eight weeks as a function of augmented, paired encounters per week, by potion composition. The combined amplifier shows a clear optimum near 3–5 encounters/week; over-dosing reduces success through tolerance and over-exposure. Placebo, oxytocin-only and dopamine-only remain at the floor (cf. Fig. 2a).

6.3 The arrow needs aiming: dose cannot substitute for contingency

Figure 3 is the central result. Sweeping contingency c (the precision with which the amplified state is paired with the specific person) against potion dose reveals a sharp structure: when c is low — the potion taken without aiming it at a paired, salient partner — P_{bond} is essentially zero *at every dose, including the maximum*. There is no dose that buys a bond without aim. Success requires *both* adequate dose *and* high contingency. This is, quantitatively, the difference between the potion and the myth’s arrow: the chemistry is only the arrowhead; contingent pairing is the aim, and an unaimed arrow — however potent — strikes nothing. It is also the formal restatement of the deepest point in the biology: a pair bond is a *conditioned* preference, so the “who” can only be supplied by pairing, never by pharmacology [48, 70].

6.4 It amplifies a spark; it cannot create one

Figure 4a sweeps baseline compatibility κ . When $\kappa \leq 0$ — a target who is indifferent or aversive — the amplifier is inert ($P_{\text{bond}} \approx 0$), no better than placebo: it cannot manufacture love from nothing or invert dislike. In the positive zone the amplifier is transformative, lifting P_{bond} from a placebo baseline near zero to 0.37 at $\kappa = 0.1$ and 0.94 at $\kappa = 0.2$. The intervention’s whole power lives in an *amplification window* above neutral valence. This both matches the evolutionary design of love as a selective system (Section 3) and carries the key ethical implication (Section 8): a realistic “love potion” can strengthen a nascent, mutual attraction but cannot coerce one into being.

6.5 Timing matters

Finally, Figure 4b confirms that pharmacokinetics impose a scheduling constraint: because the central oxytocin effect lags an intranasal dose by roughly an hour [62], P_{bond} is near zero unless

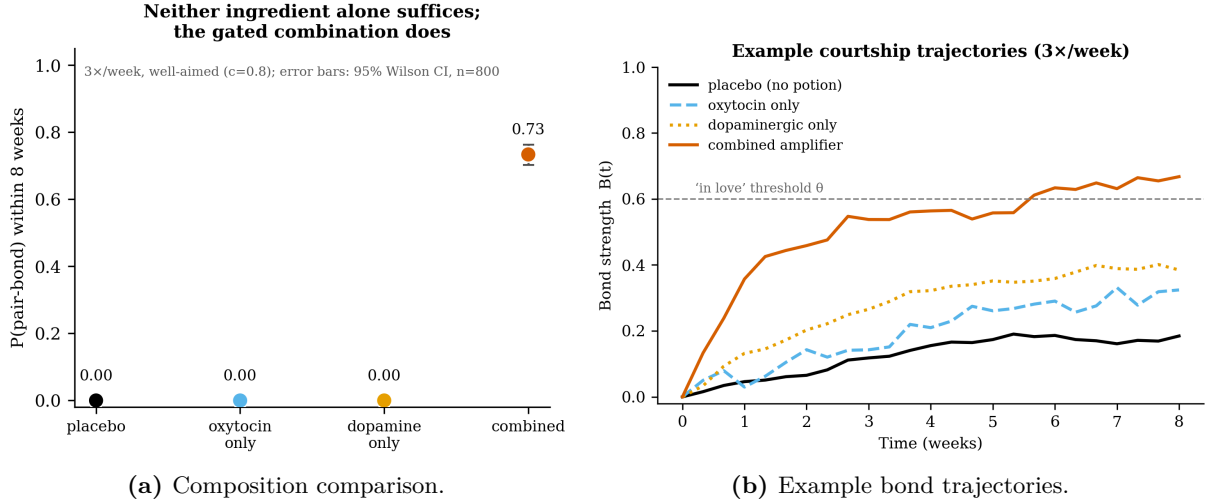


Figure 2: (a) At three well-aimed encounters/week, only the gated combination crosses the “in love” threshold; single ingredients raise bond strength but not enough. Error bars: 95% Wilson confidence intervals, $n = 800$. (b) Representative $B(t)$ trajectories under the same protocol; only the combined amplifier reaches and holds above threshold θ .

the dose–encounter timing overlap ϕ exceeds ~ 0.6 , rising to 0.79 at $\phi = 0.9$ and 0.96 at perfect timing. The arrow must not only be aimed but loosed at the right moment.

7 Part V — Can Cupid’s arrow exist? Verdict, formulation, conditions

7.1 The verdict

We can now answer the question directly. **A literal love potion — a single dose that compels a chosen person to fall durably in love against their disposition — cannot exist.** It fails on four independent grounds, each established above: it has no mechanism to specify a *target* (specificity is conditioned, not chemical; Fig. 3); it cannot invert a negative baseline (Fig. 4a); it is defeated by *pharmacokinetics* (oxytocin scarcely reaches the brain; dopamine cannot be delivered centrally and a non-specific flood can impair bonding; Section 4.2); and it is bounded by *tolerance* (Fig. 1). No refinement of a one-shot compulsion escapes these.

But a modern Cupid’s arrow — a periodically administered bonding amplifier — is feasible in principle, and would substantially raise the odds of love forming. In the model, a combined oxytocinergic–dopaminergic amplifier, dosed about three times per week, correctly timed, and paired with genuine positive interaction with one person, lifts the eight-week probability of pair-bond formation from ≈ 0 to ≈ 0.73 . This is not a magic compulsion; it is an *accelerant of an ordinary, consensual courtship*, and its power is real precisely because it works *with* the biology of love rather than against it. The romantic reading and the scientific one coincide: the potion is the arrowhead, but a person must still aim it, and the heart it strikes must already be willing to be struck.

7.2 A concrete proposed formulation

Table 2 sketches what the arrow would actually contain, mapping each component to the ingredient it serves. We stress that this is a *theoretical* formulation derived from the model and literature, not a clinical recommendation; several components (notably central oxytocin delivery) are not solved problems, and the protocol would be subject to the ethical and safety constraints of Section 8.

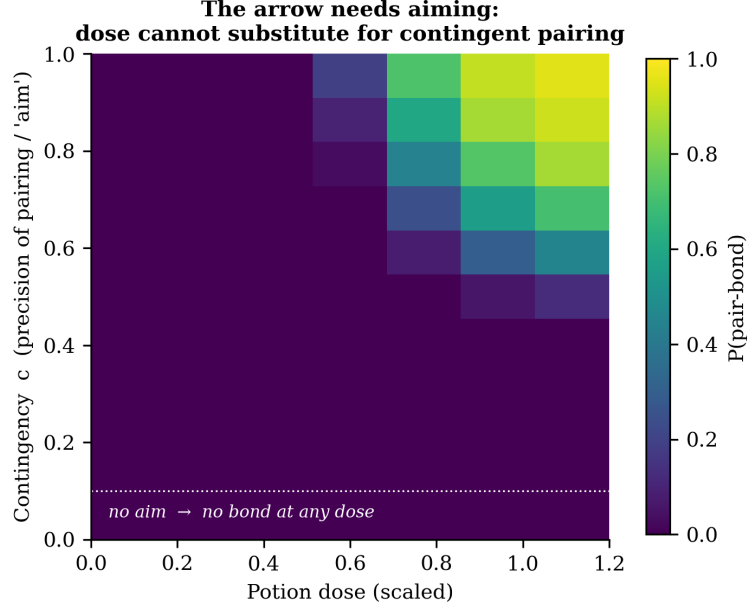


Figure 3: P_{bond} as a function of contingency c (precision of pairing) and potion dose. The entire low-contingency region is dark at all doses: without aiming, no dose produces a bond. Bonding requires the joint presence of dose and contingency.

7.3 The necessary conditions

The model makes the enabling conditions explicit. A real bonding amplifier could raise the odds of love only if *all* of the following hold:

1. **Combination, not a single agent** — reward and plasticity together (Fig. 2a).
2. **Accurate aim** — each dose paired with genuine, attentive interaction with one person; high contingency c (Fig. 3).
3. **A non-negative starting point** — some pre-existing, ideally mutual, compatibility (Fig. 4a).
4. **Periodic, moderate dosing** — repeated over weeks, near the frequency optimum, avoiding tolerance and boredom (Fig. 1).
5. **Correct timing** — doses scheduled so central effects coincide with encounters (Fig. 4b).
6. **Consent** — without which the intervention is not a love story but an assault (Section 8).

8 Ethics and limitations

Ethics. The biology that makes a coercive potion impossible also tells us what an ethical one must be. Because the amplifier only works on a non-negative baseline and only when paired with genuine positive interaction, its legitimate use is *mutual* — two consenting people choosing to deepen a bond they are already forming, much as the “love drugs” literature envisions supervised, consensual relationship support rather than conquest [22, 23, 68]. Covert administration to a non-consenting target is both the thing the myth imagines and the thing this analysis condemns: it would be a serious violation of autonomy and bodily integrity, it is what the law would rightly treat as drugging a person, and — as the model shows — it would in any case fail, because without the willing, contingent participation of the target there is no aim and no spark to amplify. The honest conclusion is therefore doubly reassuring: the dystopian love potion is *both* unethical *and* ineffective, while the version that could work is the version that requires consent.

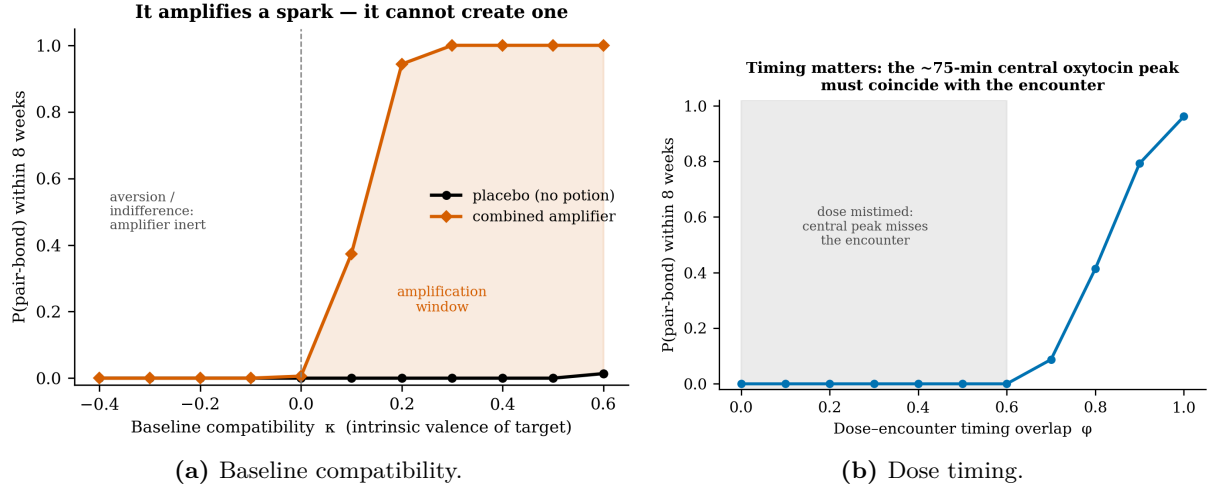


Figure 4: (a) The combined amplifier only works above neutral compatibility ($\kappa > 0$); it amplifies an existing spark and is inert against indifference or aversion. (b) The dose must be timed so the delayed central oxytocin peak coincides with the encounter; mistimed dosing ($\phi \lesssim 0.6$) is inert.

Limitations. Our model is a transparent, mechanistically-motivated abstraction, not a validated quantitative instrument. Its parameters are calibrated to the order of magnitude of the cited literature, much of which is animal data (the vole pair-bond model) or contested human data (intranasal oxytocin). The absolute probabilities should not be read as predictions for any real couple; the *qualitative structure* — combination beats single agents, contingency is necessary, compatibility gates, frequency is an inverted-U, timing matters — is the robust output, because each of those features is independently grounded in the experimental literature rather than tuned to produce a desired number. Key empirical uncertainties remain genuine open problems: whether centrally effective oxytocin delivery is achievable at all [43, 58], how far vole mechanisms generalise to humans, and whether any safe pro-reward agent exists that does not trip the flood penalty or tolerance. We have also treated “compatibility” as a scalar; real attraction is multidimensional. These are the right places for future empirical work — and the right reasons for caution before anyone mistakes a feasibility model for a recipe to be tried.

9 Conclusion

We set out to ask whether Cupid’s arrow could be real. The answer the biology gives is more interesting than a flat yes or no. Falling in love is a conditioned, dopamine-driven, oxytocin-consolidated motivational state, evolved to bind a person selectively to a suitable partner. That design forbids the fairy-tale potion — a chemical cannot choose a target, overrule aversion, or outrun its own pharmacokinetics and tolerance. But the same design leaves a real door open. Because love is built by *pairing reward and plasticity with one person over time*, a periodically administered amplifier of reward and plasticity, aimed by genuine courtship and welcomed by a willing heart, could meaningfully raise the odds that love takes hold. The modern Cupid’s arrow is not a spell that compels; it is an accelerant for a fire two people are already willing to light. It must be aimed, it must be timed, and it cannot work on a heart that is closed — which is, perhaps, exactly as it should be.

Reproducibility. The simulation (`arrow_model.py`), figure generator (`make_figures.py`) and the verified literature knowledge base accompany this manuscript. All results derive from a fixed random seed and 800 Monte-Carlo runs per condition.

Table 2: A theoretical “modern Cupid’s arrow,” by function. The behavioural components are not optional garnish — in the model they supply the contingency c without which no pharmacological component works.

Component	Candidate	Role (model ingredient)
Plasticity gate	Intranasal oxytocin, low dose (~ 8 – 24 IU), timed ~ 60 – 75 min before the encounter [57, 62]	Gates consolidation (g_O); the limiting, unsolved delivery problem
Reward / arousal	A mild, CNS-penetrant pro-reward state — pharmacologically fraught; in practice better supplied behaviourally by shared novel/arousing activity [4]	Reward magnitude (g_D), without the flood penalty of [48]
Fear/defence reduction	(Therapeutic context only) empathogenic openness, as in supervised MDMA-assisted couples settings [53]	Removes opposition to bonding
The aim (behavioural)	Contingent pairing: presence, attention, escalating self-disclosure [3], interpersonal synchrony [37], reciprocity [6]	Supplies contingency c — <i>necessary</i>
Dosing schedule	$\sim 3\times$ /week over ~ 8 weeks, timed to encounters, avoiding over-dosing	Stays at the optimum of Fig. 1

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